Data Mining for Business Analytics
MSBA 511

Logistic Regression

Logistic Regression Overview

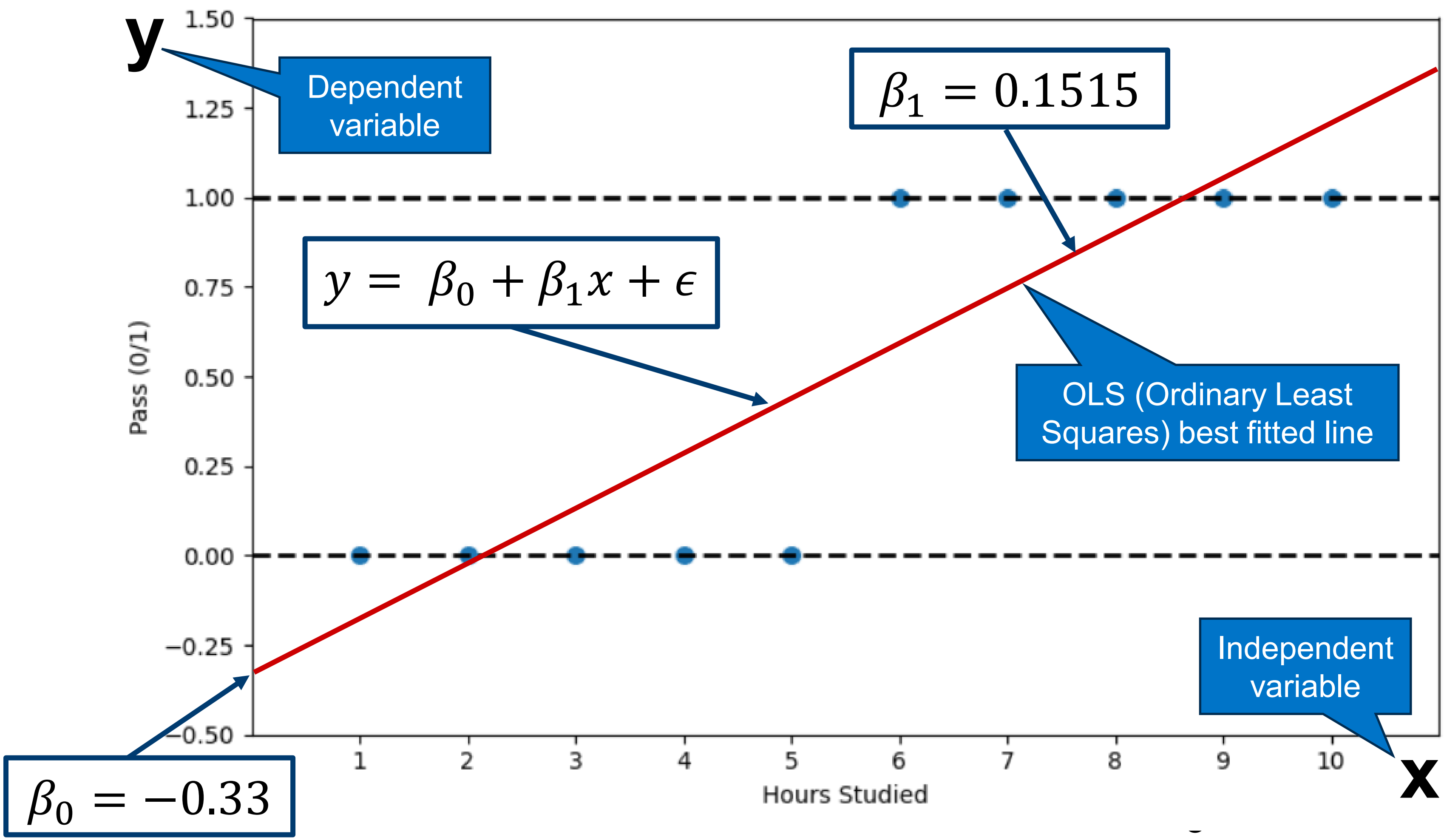
Logistic Regression is another supervised learning algorithm that extends the idea of linear regression when the outcome variable is categorical.

Some key points:

- Unlike linear regression, which predicts continuous values, logistic regression predicts the probability of belonging to a particular class.
  - ➢ Although logistic regression can be used for multi-class problems, we will focus on **binary** classification problems.
- Widely used, particularly where a structured model is useful for:
  - Profiling or finding factors that differentiate between classes.
  - Classifying a new record when the class is unknown based on the values of the predictor variables.
- Simply, fast, interpretable, and performs well with linearly separable data.

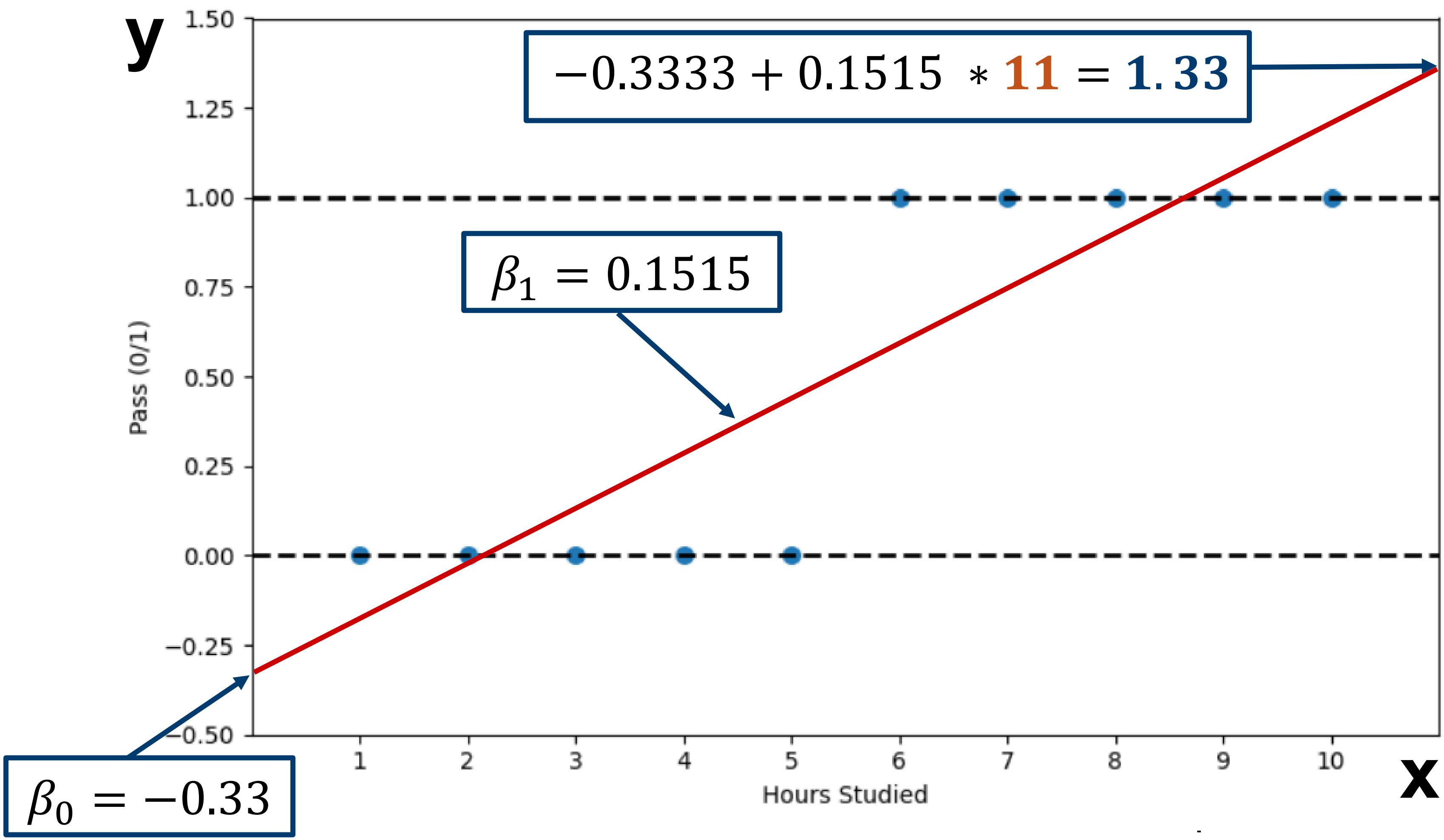
Logistic Regression Motivation

Consider the simple example of the number of hours studied and whether the student passed a certification exam.



Potential Problems

Slope is linear and predicted values can be outside of valid range. Consider the example where the student studies **11 hours**.



The Logit

Goal:

Find a function of the predictor variables that relates them to a binary outcome.

- Instead of y as outcome variable (like in linear regression), we use a function of y called the ***logit.***

- Logit can be modeled as a linear function of the predictors.

  - Mapped back to a probability and then to a class.

Step 1: The Logistic Response Function

p = probability of belonging to class 1

Standard linear function does not guarantee 0 ≤ p ≤ 1

$$p = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \ldots \beta_q x_q$$

Number of predictors

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \ldots \beta_q x_q)}}$$

Step 2: The Odds

The odds an event are defined as:

$$Odds(Y = 1) = \frac{p}{1 - p}$$

Probability of event

Or, given the odds of an event, the probability of the event can be computed by:

$$p = \frac{Odds}{1 + Odds}$$

Odds can also be related to the Predictors:

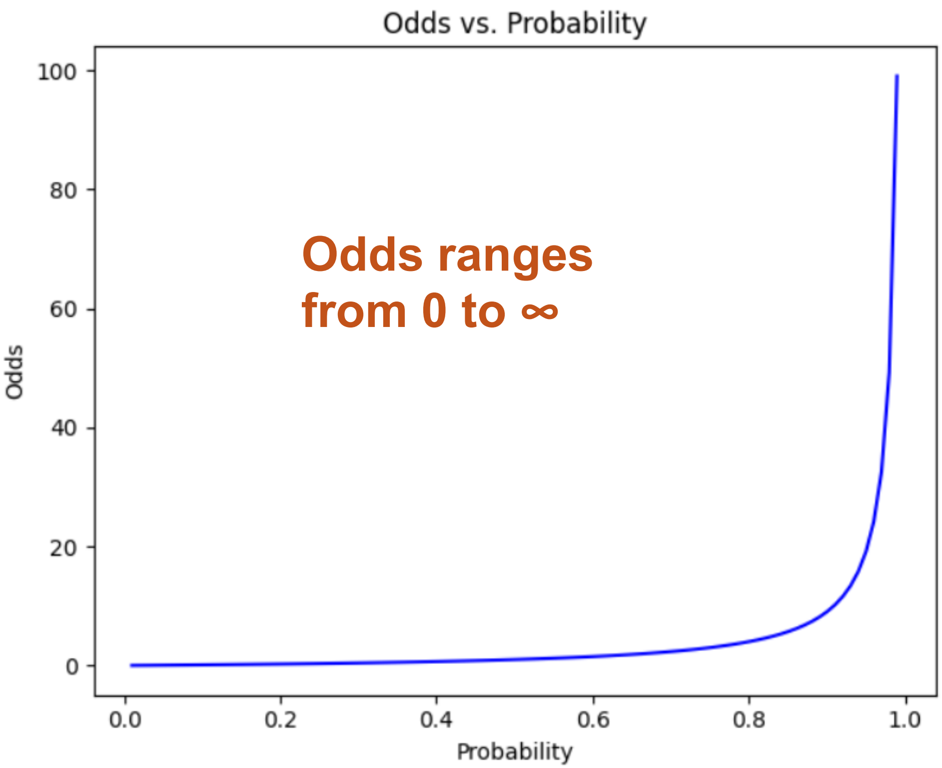
$$Odds = e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots \beta_q x_q}$$

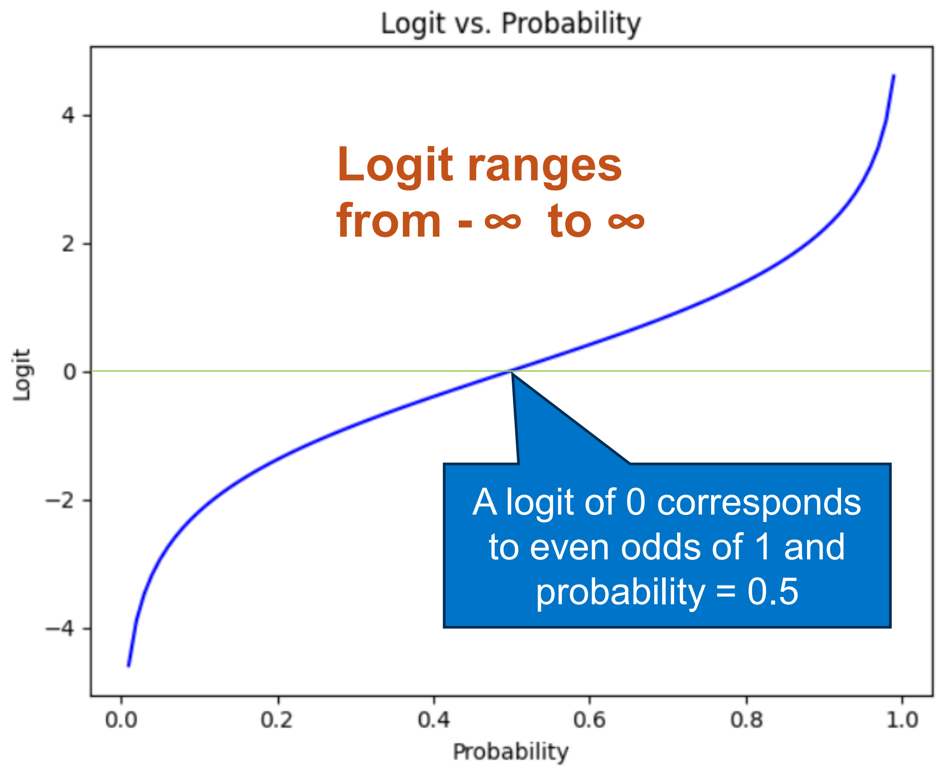
Step 3: Take Natural Log on Both Sides

This gives the standard formulation of the logistic model:

$$\log(Odds) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \ldots \beta_q x_q$$

Let's review the relationship between the probability, odds, and logit of belonging to class 1.





Let's Practice: Personal Loan Example

In this example, we will work with only 1 continuous variables and the target variable. The goal is to help a bank improve its conversion rate for personal loan acceptance in its next marketing campaign. We are working with training data for **3000** customers and the predictor variable is the customers annual income ($000s).

Assume the fitted **intercept** and **coefficient** are:

$$b_0 = -6.0627 \qquad b_1 = 0.0364$$

$$P(Personal\ Loan = Yes \mid Income = x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1)}}$$

Log(Odds)

Let's Practice: Personal Loan Example

$$b_0 = -6.0627 \qquad b_1 = 0.0364$$

$$P(Personal\ Loan = Yes \mid Income = x) = \frac{1}{1 + e^{-(-6.0627+0.0364x)}}$$

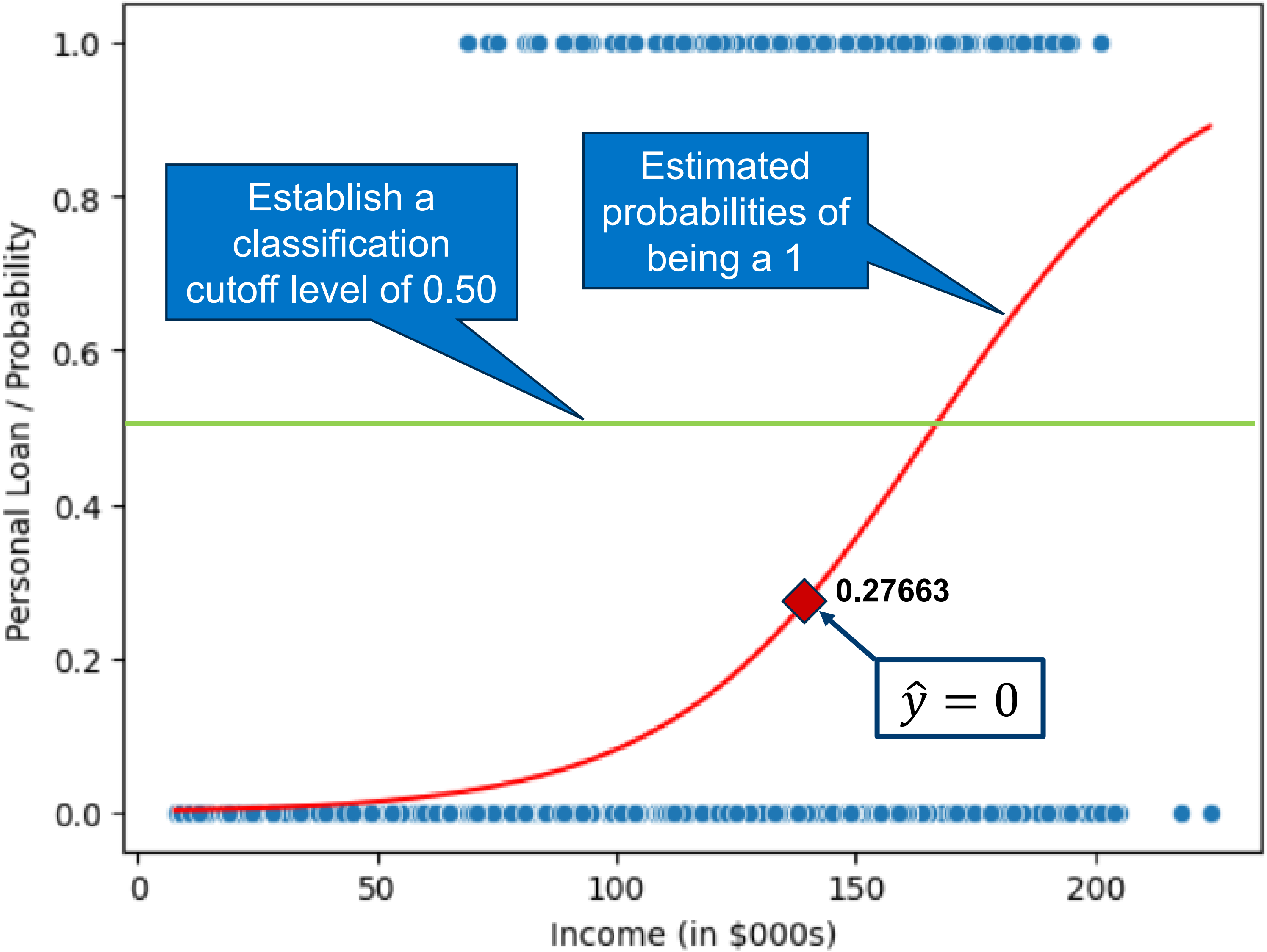
What is the probability of accepting the personal loan for a customer that makes an income of **140K**?

$$P(Personal\ Loan = Yes \mid Income = \mathbf{140}) = \frac{1}{1 + e^{-(-6.0627+0.0364*\mathbf{140})}}$$

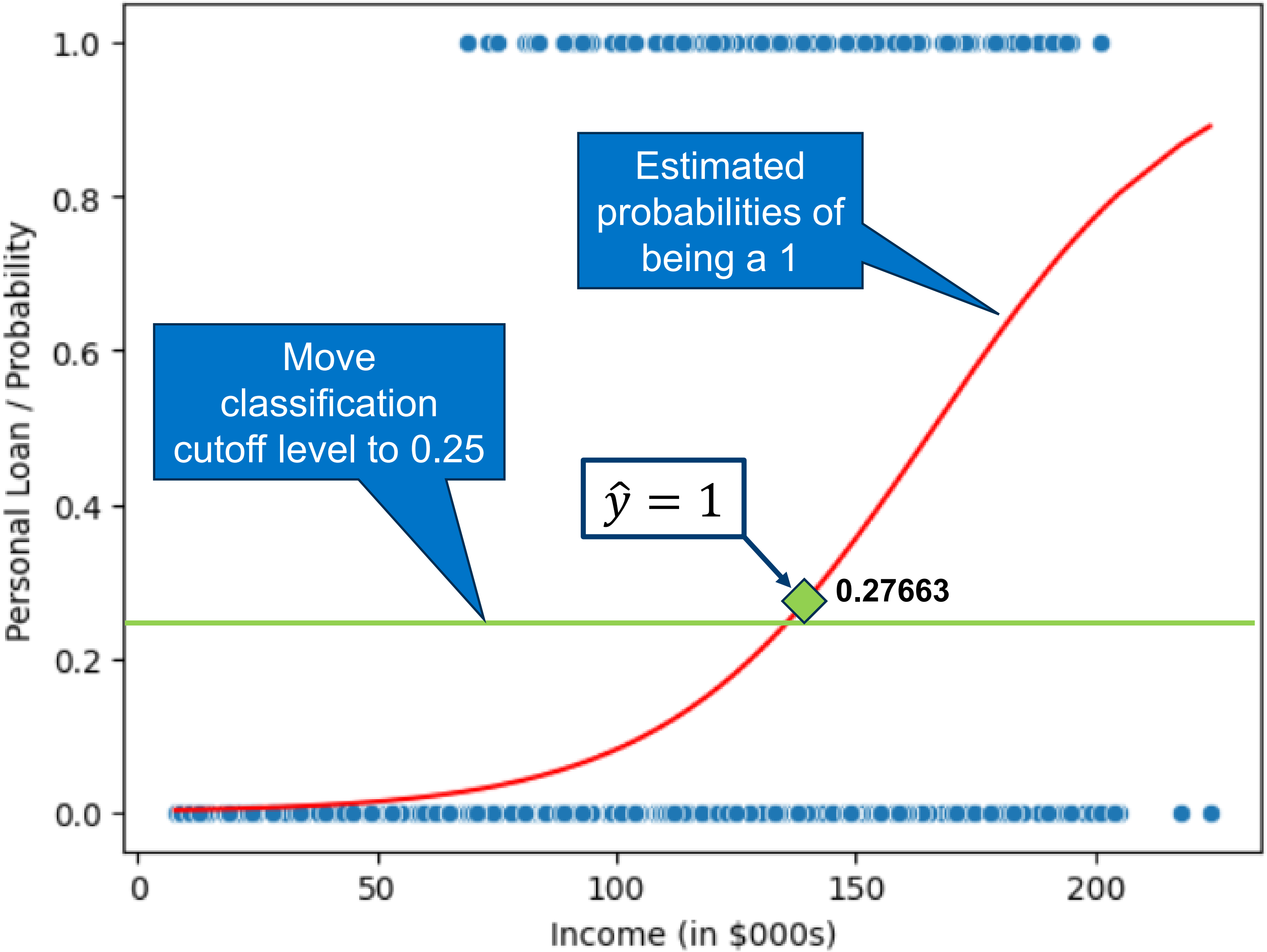
$$P(Personal\ Loan = Yes \mid Income = \mathbf{140}) = \frac{1}{1 + e^{(0.9667)}}$$

$$P(Personal\ Loan = Yes \mid Income = \mathbf{140}) = \mathbf{0.27663}$$

Visualize the Probabilities



Visualize the Probabilities



Ways to Determine Cutoff

✓ **0.50** is popular initial choice

Additional considerations:

- Maximize classification accuracy
- Maximize sensitivity (subject to min. level of specificity)
- Minimize false positives (subject to max. false negative rate)
- Minimize expected cost of misclassification (need to specify costs)

How were the Coefficients Derived?

Estimates of the coefficients (β's) are derived through an iterative process called **maximum likelihood estimation**. For our purposes, we won't go into detail for the math, but the general process is:

- Choose β values that maximize the log-likelihood.
- Optimization is done using iterative algorithms like **gradient descent** on the negative log-likelihood.

The optimal β values best separate the classes and give the most likely predictions under the logistic model.

What's with the Odds?

As we saw earlier, for predictive classification, we typically use probability with a cutoff value.

For explanatory purposes, odds have a useful interpretation:

- If we increase x_1 by one unit, holding $x_1, x_2 \ldots x_q$ constant, then b_1 is the factor by which the odds of belonging to class 1 increase.

What is Multicollinearity?

When two or more independent variables are highly correlated with each other, meaning they provide redundant information is known as multicollinearity.

Impact on Coefficients:

- Coefficients can be unstably and highly sensitive to small changes in the data.
- Can become hard to interpret the individual effect of each predictor.
- Can make variables appear statistically significant even when they are not.

Solving for Multicollinearity

Multicollinearity does not usually harm predictive accuracy; it does impact model interpretability which is important from a business sense and could lead to overfitting as well makes variables appear statistically significant even when they are not.

How to detect:

- Analyze the correlation matrix for predictors.
- Use the statistical measure Variance Inflation Factor (VIF).
  - ✓ VIF > 5 indicate potentially problematic multicollinearity.

Potential Solutions:

- Remove or combine correlated variables.
- Use regularization techniques like Lasso (L1) or Ridge (L2).
- Consider PCA to reduce dimensionality.

p-Values for Predictors

The p-value is a statistical test for the Null hypothesis that the actual coefficient (β) is equal to zero, meaning the predictor has no effect on the outcome.

Typically, you compare the p-value to some **alpha** (α) level such as 0.05.

p-Value < α:

- Suggests the variable is statistically significant and there is evidence to reject the Null hypothesis.

p-Value >= α:

- Suggests the variable may not be a useful predictor and there is evidence to not reject the Null hypothesis.

p-Values help with variable selection, but you should always consider the business objective and the tradeoff between performance and interpretability.

Selecting Subsets of Predictors

Which predictor variables should we include to accomplish the objective of finding the **simplest model** but performs sufficiently well?

There are a variety of methods for variable selection:

- **Exhaustive Search**: Evaluate every possible subset of predictors.
- **Backward Elimination**: Start with **all** and successfully eliminate least useful predictors one by one. Stop when all remaining are statistically significant.
- **Forward Selection**: Start with no predictors and add them one by one (largest contribution) and stop when the addition is not statistically significant.
- **Stepwise**: Like Forward Selection but at each step, also consider dropping non-significant predictors.

Summary

- Logistic regression is like linear regression, except that it is used with a categorical response.
- It can be used for explanatory tasks (**profiling**) or predictive tasks (**classification**).
- The predictors are related to the response y via a nonlinear function called the ***logit***.
- As in linear regression, reducing predictors can be done via variable selection.
- Logistic regression can be generalized to more than two classes.